

Yeon Seonwoo

yeon.seonwoo95@gmail.com

yeonsw.github.io

RESEARCH INTERESTS	My work focuses on building and optimizing large language models for real-world applications, with an emphasis on efficient and scalable inference, agentic search, and real-time query understanding. I have experience in training small to mid-sized language models for shopping applications and deploying medium-sized models for search systems, with a strong emphasis on latency optimization and system efficiency.	
KEYWORDS	Large Language Models (LLMs), LLM Pre-training, Inference Optimization, Agentic Search, Retrieval-Augmented Generation (RAG), Query Understanding	
EXPERIENCE	Applied Scientist, Amazon (Manhattan, NY)	Jul 2023 - Current
	• Language Model Pre-training (Stores Foundational AI): Led large-scale pre-training of small to mid-sized language models for shopping applications. • Agentic Search & LLM Efficiency/Latency Optimization (Prime Video Search): Integrated mid-sized large language models into Prime Video Search for agentic search and real-time query understanding.	
	Applied Scientist Intern, Amazon (Bellevue, WA)	Mar 2022 - Jun 2022
	Developed an unsupervised sentence representation learning method to more effectively capture semantic similarity between texts.	
	Research Scientist Intern, Adobe Research	Sep 2021 - Dec 2021
	Developed word- and relation-level semantic graph representations of documents for domain-specific document retrieval.	
EDUCATION	KAIST	Daejeon, Republic of Korea
	Ph.D. Student in School of Computing	Sep 2017 - Aug 2022
	Advisor: Alice Oh	
	Thesis title: "Distant-Supervision for Question Answering"	
	KAIST	Daejeon, Republic of Korea
	M.S. in School of Computing	Mar 2016 - Aug 2017
	Advisor: Alice Oh	
	Thesis title: "Event Prediction Using Vector Representation in Hawkes Processes"	
	Sungkyunkwan University	Suwon, Republic of Korea
	B.S. in Computer Science & Engineering	Mar 2012 - Feb 2016
AWARDS	Naver Ph.D. Fellowship, 2020	
PUBLICATION	Juhee Son, Yeon Seonwoo , Alice Oh, James Thorne, Seunghyun Yoon, "Multi-hop Database Reasoning with Virtual Knowledge Graph", <i>Workshop on Knowledge Graphs and Large Language Models 2024</i> .	
	Chani Jung, Dongkwan Kim, Jiho Jin, Jiseon Kim, Yeon Seonwoo , Yejin Choi, Alice Oh, Hyunwoo Kim, "Perceptions to Beliefs: Exploring Precursory Inferences for Theory of Mind in Large Language Models", <i>EMNLP 2024</i> .	
	Yeon Seonwoo , Guoyin Wang, Sajal Choudhary, Changmin Seo, Jiwei Li, Xiang Li, Puyang Xu, Sunghyun Park, Alice Oh. "Ranking-Enhanced Unsupervised Sentence Representation Learning." <i>ACL 2023</i>	

Yeon Seonwoo, Seunghyun Yoon, Franck Dernoncourt, Trung Bui, Alice Oh. “Virtual Knowledge Graph Construction for Zero-Shot Domain-Specific Document Retrieval.” *COLING 2022*

Changyoon Lee, **Yeon Seonwoo**, and Alice Oh. “CS1QA: A Dataset for Assisting Code-based Question Answering in an Introductory Programming Course.” *NAACL 2022*

Yeon Seonwoo^{*}, Juhee Son^{*}, Jiho Jin, Sang-Woo Lee, Ji-Hoon Kim, Jung-Woo Ha, and Alice Oh. “Two-Step Question Retrieval for Open-Domain QA.” *ACL-Findings. 2022*.

^{*}Contributed equally

Yeon Seonwoo, Sang-Woo Lee, Ji-Hoon Kim, Jung-Woo Ha, and Alice Oh. “Weakly Supervised Pre-Training for Multi-Hop Retriever.” *ACL-Findings. 2021*.
Code URL: <https://github.com/yeonsw/LOUVRE>

Yeon Seonwoo, Ji-Hoon Kim, Jung-Woo Ha, and Alice Oh. “Context-Aware Answer Extraction in Question Answering.” *EMNLP (long). 2020*.
Code URL: <https://github.com/yeonsw/BLANC>

Yeon Seonwoo, Sungjoon Park, Dongkwan Kim, and Alice Oh. “Additive Compositionality of Word Vectors.” *Workshop on Noisy User-generated Text at EMNLP. 2019*.
Code URL: <https://github.com/yeonsw/OLIVE>

Sungjoon Park, **Yeon Seonwoo**, Jiseon Kim, Jooyeon Kim and Alice Oh. “Denoising Recurrent Neural Networks for Classifying Crash-Related Events.” *IEEE Transactions on Intelligent Transportation Systems. 2019*.

Yeon Seonwoo, Sungjoon Park, and Alice Oh. “Hierarchical Dirichlet Gaussian Marked Hawkes Process for Narrative Reconstruction in Continuous Time Domain.” *EMNLP (long). 2018*.

INVITED TALKS **Semantic Alignment at Various Context Sizes**

BrainLink: State, Limitations, and Future of Large Language Models (LLM), South Korea Oct. 26, 2022

Weakly-Supervised Pre-Training for Multi-Hop Retrieval

KISTI, South Korea Feb. 08, 2022

Naver, South Korea Dec. 22, 2021

LG AI Research, South Korea Nov. 22, 2021

Context-Aware Answer Extraction in Question Answering

Naver, South Korea Jul. 01, 2020

Additive Compositionality of Word Vectors

Naver, South Korea Dec. 14, 2019

Hierarchical Dirichlet Gaussian Marked Hawkes Process for Narrative Reconstruction in Continuous Time Domain

NCSOFT AI, South Korea Jan. 25, 2019

ACADEMIC SERVICE

Reviewer

ACL Rolling Review (Oct 2021, Nov 2021, Jan 2022, Apr 2022, June 2022, July 2022, Sep 2022), NeurIPS2022, EMNLP 2022, ICLR 2022, NeurIPS 2021, EMNLP 2021, ACL 2021, ICLR 2021

SCHOLARSHIP

Korea National Science&Technology Scholarship (Mar 2012 - Feb 2016)

REFERENCES

Prof. Alice Oh, School of Computing, KAIST, alice.oh@kaist.edu